STPM Physics · Semester 1

FORMULA LIST

Mechanics & Heat · Chapters 1–12. 226 formulas with symbols, units and conditions. Numbered chapter by chapter, taken from the PhysiQ notes.

CHAPTER 1 — PHYSICAL QUANTITIES

1.1 Physical Quantities & Units

1	**Speed or velocity (dimensions)**	$[v] = [L][T]^{-1}$	[L] = length; [T] = time; SI unit m s^{-1}
2	**Acceleration (dimensions)**	$[a] = [L][T]^{-2}$	[L] = length; [T] = time; SI unit m s^{-2}
3	**Force (dimensions)**	$[F] = [M][L][T]^{-2}$	[M] = mass; [L] = length; [T] = time; SI unit N
4	**Energy and work (dimensions)**	$[M][L]^2[T]^{-2}$	SI unit J
5	**Power (dimensions)**	$[M][L]^2[T]^{-3}$	SI unit W
6	**Pressure (dimensions)**	$[M][L]^{-1}[T]^{-2}$	SI unit Pa (N m^{-2})
7	**Density (dimensions)**	$[M][L]^{-3}$	SI unit kg m^{-3}
8	**Frequency (dimensions)**	$[T]^{-1}$	SI unit Hz
9	**Momentum (dimensions)**	$[M][L][T]^{-1}$	SI unit kg m s^{-1}

1.2 Scalars and Vectors

10	**Vector components**	$A_x = A\cos\theta$; $A_y = A\sin\theta$	A = magnitude of the vector; θ = angle between A and the x-axis; A_x, A_y = components along x and y
11	**Magnitude and direction**	$A = \sqrt{A_x^2 + A_y^2}$; $\tan\theta = \frac{A_y}{A_x}$	A = magnitude; A_x, A_y = components; θ = direction angle from the x-axis
12	**Resultant of two vectors**	$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$	R = magnitude of resultant; A, B = magnitudes of the vectors; θ = angle between them (tail to tail)

13	**Resultant of perpendicular vectors**	$R = \sqrt{A^2 + B^2}$	A, B = magnitudes of the two perpendicular vectors; R = magnitude of resultant
14	**Resultant of equal vectors**	$R = 2A\cos\frac{\theta}{2}$	A = magnitude of each vector; θ = angle between them; R = magnitude of resultant
15	**Sine rule**	$\frac{a}{\sin\alpha} = \frac{b}{\sin\beta} = \frac{c}{\sin\gamma}$	a, b, c = sides of the vector triangle; α, β, γ = angles opposite a, b, c
16	**Cosine rule**	$c^2 = a^2 + b^2 - 2ab\cos\gamma$	a, b, c = sides of the vector triangle; γ = angle opposite c
17	**Resultant range**	$\lvert A - B\rvert \le R \le A + B$	A, B = magnitudes of the two vectors; R = magnitude of resultant

1.3 Uncertainties in Measurements

18	**Mean of repeated readings**	$\bar{x} = \frac{\Sigma x}{n}$	$\bar{x}$ = mean value; x = individual reading; n = number of readings
19	**Uncertainty from repeated readings**	$\Delta x = \frac{x_{max} - x_{min}}{2}$	Δx = uncertainty; x_{max}, x_{min} = largest and smallest readings
20	**Sum or difference**	$Z = A \pm B$; $\Delta Z = \Delta A + \Delta B$	ΔZ, ΔA, ΔB = absolute uncertainties, which add
21	**Product or quotient**	$Z = AB$ or $\frac{A}{B}$; $\frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$	$\frac{\Delta Z}{Z}$, $\frac{\Delta A}{A}$, $\frac{\Delta B}{B}$ = fractional uncertainties, which add
22	**Power**	$Z = A^n$; $\frac{\Delta Z}{Z} = \lvert n\rvert\frac{\Delta A}{A}$	n = power; $\frac{\Delta Z}{Z}$, $\frac{\Delta A}{A}$ = fractional uncertainties
23	**General combination**	$Z = A^m\frac{B^n}{C^p}$; $\frac{\Delta Z}{Z} = \frac{\lvert m\rvert\Delta A}{A} + \frac{\lvert n\rvert\Delta B}{B} + \frac{\lvert p\rvert\Delta C}{C}$	m, n, p = powers; $\frac{\Delta A}{A}$, $\frac{\Delta B}{B}$, $\frac{\Delta C}{C}$ = fractional uncertainties
24	**Constant factor**	$Z = kA$; $\Delta Z = k\Delta A$	k = exact number; it does not add to the fractional uncertainty

CHAPTER 2 — KINEMATICS

2.1 Linear Motion

1	**Velocity–time equation**	$v = u + at$	u = initial velocity; v = final velocity (m s^{-1}); a = constant acceleration (m s^{-2}); t = time (s); constant acceleration in a straight line only

2	**Displacement equation**	$s = ut + \frac{1}{2}at^2$	s = displacement (m); u = initial velocity (m s^{-1}); a = constant acceleration (m s^{-2}); t = time (s)
3	**Velocity–displacement equation**	$v^2 = u^2 + 2as$	u = initial velocity; v = final velocity (m s^{-1}); a = constant acceleration (m s^{-2}); s = displacement (m)
4	**Displacement from mean velocity**	$s = \frac{1}{2}(u + v)t$	s = displacement (m); u = initial velocity; v = final velocity (m s^{-1}); t = time (s)
5	**Displacement from final velocity**	$s = vt - \frac{1}{2}at^2$	s = displacement (m); v = final velocity (m s^{-1}); a = constant acceleration (m s^{-2}); t = time (s); valid for constant acceleration in a straight line only
6	**Average velocity**	$\frac{\Delta s}{\Delta t}$	Δs = change in displacement (m); Δt = time interval (s)
7	**Average speed**	total distance/total time	distance in m; time in s; speed in m s^{-1}
8	**Vertical motion**	$a = -g = -9.81\ m\ s^{-2}$	Take one direction positive (e.g. upward); g = gravitational acceleration; at the highest point v = 0

2.2 Projectiles

9	**Launch velocity components**	$u_x = u\cos\theta$; $u_y = u\sin\theta$	u = launch speed (m s^{-1}); θ = angle above the horizontal; u_x, u_y = horizontal and vertical components
10	**Velocity and position at time t**	$v_x = u\cos\theta$; $v_y = u\sin\theta - gt$; $x = (u\cos\theta)t$; $y = (u\sin\theta)t - \frac{1}{2}gt^2$	v_x, v_y = velocity components (m s^{-1}); x, y = horizontal and vertical displacements (m); g = gravitational acceleration (m s^{-2}); t = time (s)
11	**Vertical velocity–displacement**	$v_y^2 = u^2\sin^2\theta - 2gy$	v_y = vertical velocity (m s^{-1}); y = vertical displacement (m); u = launch speed (m s^{-1}); g = gravitational acceleration (m s^{-2})
12	**Speed**	$v = \sqrt{v_x^2 + v_y^2}$	v = speed (m s^{-1}); v_x, v_y = velocity components (m s^{-1})
13	**Angle below horizontal**	$\varphi = \tan^{-1}\frac{\lvert v_y\rvert}{v_x}$	φ = angle of velocity below the horizontal; v_x, v_y = velocity components (m s^{-1})
14	**Time to maximum height**	$t = u\,\frac{\sin\theta}{g}$	t = time (s); u = launch speed (m s^{-1}); θ = launch angle; g = gravitational acceleration (m s^{-2})

15	**Time of flight**	$T = 2u\,\frac{\sin\theta}{g}$	T = time of flight (s); u = launch speed (m s^{-1}); θ = launch angle; g = gravitational acceleration (m s^{-2}); landing at the same level
16	**Maximum height**	$H = \frac{u^2\sin^2\theta}{2g}$	H = maximum height (m); u = launch speed (m s^{-1}); θ = launch angle; g = gravitational acceleration (m s^{-2})
17	**Range**	$R = \frac{u^2\sin 2\theta}{g}$	R = horizontal range (m); u = launch speed (m s^{-1}); θ = launch angle; R is maximum at θ = 45°, and θ and (90° − θ) give the same range
18	**Trajectory**	$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$	x, y = horizontal and vertical displacements (m); u = launch speed (m s^{-1}); θ = launch angle; g = gravitational acceleration (m s^{-2})
19	**Horizontal launch from height h**	$t = \sqrt{\frac{2h}{g}}$; $x = ut$	t = time of fall (s); h = height (m); x = range (m); u = horizontal launch speed (m s^{-1}); g = 9.81 m s^{-2}

CHAPTER 3 — DYNAMICS

3.1 Newton's Laws of Motion

1	**Newton's second law**	$F = ma$	F = ΣF = resultant force (N); m = mass (kg), constant; a = acceleration (m s^{-2})
2	**Constant-mass form of the law**	$F = \frac{mv - mu}{t}$	F = force (N); m = mass (kg); u, v = initial and final velocities (m s^{-1}); t = time (s)
3	**Lift accelerating upward**	$R = m(g + a)$	R = reaction force on the person (N); m = mass (kg); g = gravitational acceleration (m s^{-2}); a = acceleration of lift (m s^{-2})
4	**Lift accelerating downward**	$R = m(g - a)$	R = reaction force on the person (N); m = mass (kg); g = gravitational acceleration (m s^{-2}); a = acceleration of lift (m s^{-2})
5	**Free fall of the lift**	$R = 0$	R = reaction force (N); apparent weightlessness
6	**Connected bodies, common acceleration**	$a = \frac{(m_1 - m_2)g}{m_1 + m_2}$	a = common acceleration (m s^{-2}); m_1, m_2 = masses (kg); g = gravitational acceleration (m s^{-2}); light string over a smooth pulley

7	**Tension in connecting string**	$T = \frac{2m_1m_2g}{m_1 + m_2}$	T = tension (N); m_1, m_2 = masses (kg); g = gravitational acceleration (m s^{-2}); light string over a smooth pulley
8	**Smooth incline**	$a = g \sin\theta$; $N = mg \cos\theta$	a = acceleration down the slope (m s^{-2}); θ = angle of incline; N = normal reaction (N); m = mass (kg); g = gravitational acceleration (m s^{-2})

3.2 Linear Momentum

9	**Linear momentum**	$p = mv$	p = momentum (kg m s^{-1}); m = mass (kg); v = velocity (m s^{-1})
10	**Conservation of momentum**	$\Sigma p(\text{before}) = \Sigma p(\text{after})$; $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$	m_1, m_2 = masses (kg); u_1, u_2 = initial velocities and v_1, v_2 = final velocities (m s^{-1}); one direction taken positive
11	**Impulse (constant force)**	$J = F\Delta t = mv - mu = \Delta p$	J = impulse (N s); F = constant force (N); Δt = time (s); u, v = initial and final velocities (m s^{-1}); Δp = change in momentum (kg m s^{-1})
12	**Impulse (varying force)**	$J = \int F\,dt$	J = impulse (N s) = area under the F–t graph; F = force (N); t = time (s)
13	**Force as rate of change of momentum**	$F = \frac{\Delta p}{\Delta t} = \frac{mv - mu}{\Delta t}$	F = force (N); Δp = change in momentum (kg m s^{-1}); Δt = time interval (s); m = mass (kg); u, v = initial and final velocities (m s^{-1})
14	**Continuous stream hitting a surface**	$F = \frac{dm}{dt}(v - u)$	F = force (N); $\frac{dm}{dt}$ = mass flow rate (kg s^{-1}); u, v = velocities before and after (m s^{-1})
15	**Rebound from a wall**	$\Delta p = m(v + u)$	Δp = magnitude of change in momentum (kg m s^{-1}); m = mass (kg); u = speed of approach; v = speed of return (m s^{-1})
16	**Explosion or recoil from rest**	$m_1v_1 + m_2v_2 = 0$; $v_2 = -\frac{m_1v_1}{m_2}$	m_1, m_2 = masses (kg); v_1, v_2 = velocities after the explosion (m s^{-1})
17	**Kinetic energy and momentum**	$E_k = \frac{p^2}{2m}$	E_k = kinetic energy (J); p = momentum (kg m s^{-1}); m = mass (kg)

3.3 Elastic and Inelastic Collisions

18	**Conservation of momentum**	$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$	m_1, m_2 = masses (kg); u_1, u_2 = velocities before collision (m s^{-1}); v_1, v_2 = velocities after collision (m s^{-1}); holds for both elastic and inelastic collisions
19	**Elastic collision: kinetic energy**	$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$	Elastic collisions only; total kinetic energy (J) is conserved

20	**Elastic collision: velocities**	$u_1 - u_2 = v_2 - v_1$	Elastic collisions only; u = velocities before, v = velocities after (m s^{-1})
21	**Perfectly inelastic collision**	$m_1u_1 + m_2u_2 = (m_1 + m_2)v$	Perfectly inelastic collision; v = common velocity after collision (m s^{-1}); masses in kg
22	**Elastic, m_2 initially at rest**	$v_1 = \frac{(m_1 - m_2)u_1}{m_1 + m_2}$; $v_2 = \frac{2m_1u_1}{m_1 + m_2}$	Elastic collision with m_2 initially at rest; v_1, v_2 = final velocities (m s^{-1}); u_1 = initial velocity of m_1 (m s^{-1})
23	**Loss of kinetic energy**	Loss = E_k(before) − E_k(after); fractional loss = $\frac{\Delta E_k}{E_k(\text{before})}$	E_k = kinetic energy (J); ΔE_k = loss of kinetic energy (J)
24	**Perfectly inelastic, m_2 at rest**	$\frac{\Delta E_k}{E_k} = \frac{m_2}{m_1 + m_2}$	Perfectly inelastic collision with m_2 at rest; $\frac{\Delta E_k}{E_k}$ = fractional loss of kinetic energy
25	**Two-dimensional collisions**	Σp_x before = Σp_x after; Σp_y before = Σp_y after	p = momentum (kg m s^{-1}); conservation applied along x and along y separately

3.4 Friction

26	**Static friction**	$F_s \le \mu_s N$; limiting friction $F_{max} = \mu_s N$	F_s = static friction (N); μ_s = coefficient of static friction; N = normal reaction (N); F_{max} = limiting friction (N)
27	**Kinetic friction**	$F_k = \mu_k N$	F_k = kinetic friction (N); μ_k = coefficient of kinetic friction; N = normal reaction (N)
28	**Normal reaction, pulled block**	$N = mg - P \sin\theta$	Block on horizontal surface pulled by force P at angle θ above the horizontal; m = mass (kg); g = acceleration of free fall (m s^{-2}); P = pulling force (N)
29	**Horizontal motion, pulled block**	$\Sigma F_x = P\cos\theta - \mu N = ma$	Block on horizontal surface pulled by force P at angle θ above the horizontal; μ = coefficient of friction; a = acceleration (m s^{-2})
30	**Normal reaction on incline**	$N = mg\cos\theta$	Incline of angle θ; m = mass (kg); g = acceleration of free fall (m s^{-2})
31	**Weight component down slope**	$mg\sin\theta$	Incline of angle θ; component of weight down the slope (N)
32	**Point of sliding (limiting angle)**	$\tan\theta = \mu_s$	On the point of sliding; θ = angle of friction; μ_s = coefficient of static friction

33	**Sliding at constant speed**	$\tan\theta = \mu_k$	Sliding down the incline at constant speed; μ_k = coefficient of kinetic friction
34	**Acceleration down rough incline**	$a = g(\sin\theta - \mu_k \cos\theta)$	a = acceleration down the slope (m s^{-2}); g = acceleration of free fall (m s^{-2}); θ = angle of incline; μ_k = coefficient of kinetic friction
35	**Acceleration up slope when pushed**	$a = -g(\sin\theta + \mu_k \cos\theta)$	Body pushed up the slope; negative sign indicates deceleration (m s^{-2}); other symbols as above
36	**Stopping distance on rough level surface**	$s = \frac{u^2}{2\mu g}$	Body sliding on a rough level surface; s = stopping distance (m); u = initial speed (m s^{-1}); μ = coefficient of friction; g = acceleration of free fall (m s^{-2})
37	**Deceleration on rough level surface**	$a = \mu g$	Body sliding on a rough level surface; a = deceleration (m s^{-2})

CHAPTER 4 — WORK, ENERGY AND POWER

4.1 Work Done by a Force

1	**Work done by a constant force**	$W = Fs\cos\theta$	Constant force; W = work done (J); F = force (N); s = displacement (m); θ = angle between force and displacement
2	**Work done against gravity**	$W = mgh$	Raising a mass m (kg) through height h (m); g = acceleration of free fall (m s^{-2}); independent of the path
3	**Work done on a spring**	$W = \frac{1}{2}kx^2$ = area under the F–x graph	Spring obeying Hooke's law F = kx; work done in stretching; k = force constant (N m^{-1}); x = extension (m)
4	**Net work (work–energy)**	$W_{net} = \Sigma F \cdot s = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$	W_{net} = net work done (J); m = mass (kg); u = initial speed (m s^{-1}); v = final speed (m s^{-1})
5	**Work done against friction**	$W = \mu mgs$	Level surface; μ = coefficient of friction; m = mass (kg); g = acceleration of free fall (m s^{-2}); s = distance moved (m)

4.2 Conservation of Energy

6	**Conservation of mechanical energy**	$\frac{1}{2}mu^2 + mgh_1 = \frac{1}{2}mv^2 + mgh_2$	No resistive force; m = mass (kg); u, v = initial and final speeds (m s^{-1}); h_1, h_2 = initial and final heights (m); g = acceleration of free fall (m s^{-2})

7	**Energy with friction**	$\frac{1}{2}mu^2 + mgh_1 = \frac{1}{2}mv^2 + mgh_2 + F_f d$	Initial energy = final energy + work done against friction; F_f = frictional force (N); d = distance moved against friction (m)
8	**Spring released from natural length**	$\frac{1}{2}kx^2 = \frac{1}{2}mv^2$	Spring and mass released from the natural length; k = force constant (N m^{-1}); x = extension (m); m = mass (kg); v = speed (m s^{-1})
9	**Spring and mass, vertical motion**	$\frac{1}{2}kx^2 + mgh$	Energy interchange between the elastic and gravitational terms for vertical motion
10	**Simple pendulum speed**	$v = \sqrt{2gL(1 - \cos\theta)}$	Simple pendulum released from angle θ; v = speed at the lowest point (m s^{-1}); L = length (m); g = acceleration of free fall (m s^{-2})
11	**Loss of energy**	Loss of energy = $E_{initial} - E_{final}$	E = energy (J)
12	**Work–energy principle**	Net work = ΔE_k	ΔE_k = change in kinetic energy (J)

4.3 Power and Efficiency

13	**Power**	$P = \frac{W}{t} = \frac{\Delta E}{t}$	P = power (W); W = work done (J); ΔE = energy transferred (J); t = time (s)
14	**Power and force**	$P = Fv$	F and v in the same direction; F = driving force (N); v = velocity (m s^{-1})
15	**Power to raise a load**	$P = mgv$; $P = \frac{mgh}{t}$	Raising a load at constant speed; m = mass (kg); g = acceleration of free fall (m s^{-2}); v = speed (m s^{-1}); h = height (m); t = time (s)
16	**Efficiency**	η = (useful energy output/energy input) × 100% = (useful power output/power input) × 100%	η = efficiency (%); energies in J; powers in W
17	**Power balance**	Input power = output power + power wasted	Powers in W
18	**Vehicle on a slope**	F = resistive force + mg sin θ; P = Fv	Vehicle at constant speed on a slope of angle θ; F = driving force (N); m = mass (kg); g = acceleration of free fall (m s^{-2}); v = speed (m s^{-1})
19	**Power of flowing fluid**	$P = \frac{1}{2}\frac{dm}{dt}v^2 = \frac{1}{2}\rho A v^3$	Flowing fluid (e.g. water jet or wind); $\frac{dm}{dt}$ = mass flow rate (kg s^{-1}); v = speed (m s^{-1}); ρ = density (kg m^{-3}); A = cross-sectional area (m^2)

CHAPTER 5 — CIRCULAR MOTION

5.1 Angular Velocity and Centripetal Acceleration

1	**Angular displacement**	$\theta = \frac{s}{r}$	θ = angle (rad); s = arc length (m); r = radius (m); 360° = 2π rad
2	**Angular velocity**	$\omega = \frac{d\theta}{dt} = \frac{2\pi}{T} = 2\pi f$	ω = angular velocity (rad s^{-1}); T = period (s); f = frequency (Hz)
3	**Linear speed in circular motion**	$v = r\omega$	v = linear (tangential) speed (m s^{-1}); r = radius (m); ω = angular velocity (rad s^{-1})
4	**Centripetal acceleration**	$a = \frac{v^2}{r} = r\omega^2 = v\omega$	a = centripetal acceleration (m s^{-2}), directed towards the centre; v = speed (m s^{-1}); r = radius (m); ω = angular velocity (rad s^{-1})
5	**Angular velocity (constant α)**	$\omega = \omega_0 + \alpha t$	ω_0 = initial angular velocity; ω = final angular velocity (rad s^{-1}); α = constant angular acceleration (rad s^{-2}); t = time (s)
6	**Angular displacement (constant α)**	$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$	θ = angular displacement (rad); ω_0 = initial angular velocity (rad s^{-1}); α = constant angular acceleration (rad s^{-2}); t = time (s)
7	**Angular velocity–displacement (constant α)**	$\omega^2 = \omega_0^2 + 2\alpha\theta$	ω_0, ω = initial and final angular velocities (rad s^{-1}); α = constant angular acceleration (rad s^{-2}); θ = angular displacement (rad)
8	**Angular displacement from mean ω (constant α)**	$\theta = \frac{1}{2}(\omega_0 + \omega)t$	θ = angular displacement (rad); ω_0, ω = initial and final angular velocities (rad s^{-1}); t = time (s); rotation with constant angular acceleration

5.2 Centripetal Force

9	**Centripetal force**	$F = \frac{mv^2}{r} = mr\omega^2 = mv\omega$	F = centripetal force (N); m = mass (kg); v = speed (m s^{-1}); r = radius (m); ω = angular velocity (rad s^{-1})
10	**Horizontal circle on a string / conical pendulum**	$T\sin\theta = \frac{mv^2}{r}$; $T\cos\theta = mg$; $\tan\theta = \frac{v^2}{rg}$	T = tension (N); θ = angle of string to the vertical; m = mass (kg); v = speed (m s^{-1}); r = radius (m); g = gravitational acceleration (m s^{-2})
11	**Level bend (friction provides F)**	$\mu mg \geq \frac{mv^2}{r}$; $v_{max} = \sqrt{\mu gr}$	μ = coefficient of friction; m = mass (kg); v_{max} = maximum speed without skidding (m s^{-1}); r = radius of bend (m)

12	**Banked bend, no friction**	$\tan\theta = \frac{v^2}{rg}$	θ = angle of banking; v = speed (m s^{-1}); r = radius (m); g = gravitational acceleration (m s^{-2})
13	**Vertical circle, top**	$T + mg = \frac{mv^2}{r}$	T = tension (N); m = mass (kg); v = speed at the top (m s^{-1}); r = radius (m)
14	**Vertical circle, bottom**	$T - mg = \frac{mv^2}{r}$	T = tension (N); m = mass (kg); v = speed at the bottom (m s^{-1}); r = radius (m)
15	**Minimum speed at top of vertical circle**	$v = \sqrt{gr}$	Condition T = 0 at the top; v = minimum speed (m s^{-1}); r = radius (m); g = gravitational acceleration (m s^{-2})
16	**Top of a convex bridge**	$mg - N = \frac{mv^2}{r}$	N = normal reaction (N); m = mass (kg); v = speed (m s^{-1}); r = radius of curvature (m); loop or bridge
17	**Bottom of a dip**	$N - mg = \frac{mv^2}{r}$	N = normal reaction (N); m = mass (kg); v = speed (m s^{-1}); r = radius of curvature (m); loop or bridge

CHAPTER 6 — GRAVITATION

6.1 Newton's Law of Gravitation

1	**Newton's law of gravitation**	$F = \frac{GMm}{r^2}$	F = gravitational force (N); G = 6.67×10^{-11} N m^2 kg^{-2}; M, m = masses (kg); r = separation of the centres (m)
2	**Surface gravitational field strength**	$g = \frac{GM}{R^2}$; $GM = gR^2$	g = gravitational field strength at the surface (N kg^{-1}); M = mass of planet (kg); R = radius of planet (m)
3	**Planet of density ρ**	$M = \frac{4}{3}\pi R^3\rho$; $g = \frac{4}{3}\pi G\rho R$	M = mass (kg); R = radius (m); ρ = density (kg m^{-3}); G = gravitational constant
4	**Circular orbit**	$\frac{GMm}{r^2} = \frac{mv^2}{r} = mr\omega^2$; $T^2 = \frac{4\pi^2}{GM}r^3$	M = mass of central body (kg); m = orbiting mass (kg); r = orbital radius (m); v = orbital speed (m s^{-1}); ω = angular velocity (rad s^{-1}); T = period (s)
5	**Kepler's third law**	$\frac{T_1^2}{T_2^2} = \frac{r_1^3}{r_2^3}$	T_1, T_2 = periods (s); r_1, r_2 = orbital radii (m); two orbits about the same body
6	**Mass of the central body**	$M = \frac{4\pi^2r^3}{GT^2}$	M = mass of central body (kg); r = orbital radius (m); T = period (s); G = gravitational constant

7	**Density method**	$M = \rho\frac{4}{3}\pi R^3$; $g \propto R$	M = mass (kg); ρ = density (kg m^{-3}); R = radius (m); $g \propto R$ holds for equal density

6.2 Gravitational Field and Potential

8	**Gravitational field strength**	$g = \frac{GM}{r^2}$	g = field strength (N kg^{-1}); M = mass of the source (kg); r = distance from centre (m); valid for $r \geq R$ (outside a spherical body)
9	**Gravitational potential**	$V = -\frac{GM}{r}$	V = gravitational potential (J kg^{-1}); M = mass of source (kg); r = distance from centre (m)
10	**Gravitational potential energy**	$U = -\frac{GMm}{r}$	U = potential energy (J); M = mass of source (kg); m = mass in the field (kg); r = distance from centre (m)
11	**Field and potential gradient**	$g = -\frac{dV}{dr}$	g = field strength (N kg^{-1}); V = potential (J kg^{-1}); r = distance (m)
12	**Change in potential energy**	$\Delta U = m(V_2 - V_1) = GMm(\frac{1}{r_1} - \frac{1}{r_2})$	ΔU = change in potential energy (J); V_1, V_2 = potentials (J kg^{-1}) at r_1, r_2 (m)
13	**Change in potential energy near surface**	$\Delta U = mg\Delta h$	m = mass (kg); g = gravitational field strength (N kg^{-1}); Δh = change in height (m); near the surface
14	**Escape speed**	$\frac{1}{2}mv^2 - \frac{GMm}{R} = 0$; $v_{esc} = \sqrt{\frac{2GM}{R}} = \sqrt{2gR}$	v_{esc} = escape speed (m s^{-1}); M = mass of planet (kg); R = radius (m); g = surface field strength (N kg^{-1}); for the Earth about 1.1×10^4 m s^{-1}
15	**Energy conservation between two points**	$\frac{1}{2}mv_1^2 - \frac{GMm}{r_1} = \frac{1}{2}mv_2^2 - \frac{GMm}{r_2}$	v_1, v_2 = speeds (m s^{-1}) at distances r_1, r_2 (m); m = mass (kg); M = mass of source (kg)
16	**Several masses**	$V = \Sigma V$	V = total potential (J kg^{-1}); scalar sum of the potentials due to each mass

6.3 Satellite Motion

17	**Orbital speed**	$v = \sqrt{\frac{GM}{r}}$	v = orbital speed (m s^{-1}); M = mass of central body (kg); r = R + h = orbital radius (m)
18	**Period and mass of central body**	$T = \frac{2\pi r}{v}$; $T^2 = \frac{4\pi^2}{GM}r^3$; $M = \frac{4\pi^2 r^3}{GT^2}$	T = period (s); r = orbital radius (m); v = orbital speed (m s^{-1}); M = mass of central body (kg)

19	**Angular velocity and acceleration**	$\omega = \sqrt{\frac{GM}{r^3}}$; $a = g_r = \frac{GM}{r^2}$	ω = angular velocity (rad s^{-1}); a = centripetal acceleration = field strength g_r at radius r (m s^{-2})
20	**Kinetic energy of satellite**	$K = \frac{GMm}{2r}$	K = kinetic energy (J); m = satellite mass (kg); M = mass of central body (kg); r = orbital radius (m)
21	**Potential energy of satellite**	$U = -\frac{GMm}{r}$	U = potential energy (J); m = satellite mass (kg); M = mass of central body (kg); r = orbital radius (m)
22	**Total energy of satellite**	$E = K + U = -\frac{GMm}{2r}$; $E = -K$; $U = -2K$	E = total energy (J); K = kinetic energy (J); U = potential energy (J)
23	**Orbital speed near the surface**	$v = \sqrt{gR} \approx 7.9\times10^3$ m s^{-1}	v = orbital speed (m s^{-1}); g = surface field strength (N kg^{-1}); R = radius of planet (m)
24	**Escape speed from orbital speed**	$v_{esc} = \sqrt{2}\ v_{orbit}$	v_{esc} = escape speed (m s^{-1}); v_{orbit} = orbital speed (m s^{-1})
25	**Energy to raise a satellite**	$\Delta E = \frac{GMm}{2}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$	ΔE = energy needed (J); r_1, r_2 = initial and final orbital radii (m); m = satellite mass (kg); M = mass of central body (kg)
26	**Geostationary radius**	$r^3 = \frac{GMT^2}{4\pi^2}$	r = orbital radius (m); T = 24 h = 8.64×10^4 s; M = mass of Earth (kg)

CHAPTER 7 — ROTATION OF RIGID BODIES

7.1 Torque and Angular Acceleration

1	**Torque of a force**	$\tau = Fr\sin\theta = F_\perp r = Fd$	τ = torque (N m); F = force (N); r = distance from axis (m); θ = angle between F and r; $F_\perp$ = component of F perpendicular to r; $d = r\sin\theta$ = perpendicular distance (m)
2	**Couple**	$\tau = Fd$	τ = torque of the couple (N m); F = magnitude of each force (N); d = perpendicular separation of the two forces (m)
3	**Torque and angular acceleration**	$\tau = I\alpha$	τ = torque (N m); I = moment of inertia (kg m^2); α = angular acceleration (rad s^{-2})

4	**Linear–rotational links**	$s = r\theta$; $v = r\omega$; $a_t = r\alpha$	s = arc length (m); v = tangential speed (m s^{-1}); a_t = tangential acceleration (m s^{-2}); θ = angle (rad); ω = angular velocity (rad s^{-1}); α = angular acceleration (rad s^{-2}); r = radius (m)
5	**Equations of rotational motion**	$\omega = \omega_0 + \alpha t$; $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$; $\omega^2 = \omega_0^2 + 2\alpha\theta$	ω_0, ω = initial and final angular velocities (rad s^{-1}); θ = angular displacement (rad); t = time (s); constant α (rad s^{-2})
6	**Wheel or pulley with a hanging mass**	$mg - T = ma$; $Tr = I\alpha$; $a = r\alpha$	m = hanging mass (kg); T = tension (N); a = linear acceleration of mass (m s^{-2}); r = radius of wheel (m); I = moment of inertia (kg m^2); α = angular acceleration (rad s^{-2})

7.2 Moment of Inertia

7	**Point mass**	$I = mr^2$	I = moment of inertia (kg m^2); m = mass (kg); r = perpendicular distance from the axis (m)
8	**System of masses**	$I = \Sigma m_i r_i^2 = m_1 r_1^2 + m_2 r_2^2 + \ldots$	I = moment of inertia (kg m^2); m_i = mass of the i-th particle (kg); r_i = its perpendicular distance from the axis (m)
9	**Thin hoop or ring**	$I = MR^2$	M = mass (kg); R = radius (m); about its axis
10	**Solid disc or cylinder**	$I = \frac{1}{2}MR^2$	M = mass (kg); R = radius (m); about its axis
11	**Solid sphere**	$I = \frac{2}{5}MR^2$	M = mass (kg); R = radius (m); about a diameter
12	**Thin spherical shell**	$I = \frac{2}{3}MR^2$	M = mass (kg); R = radius (m)
13	**Thin rod about its centre**	$I = \frac{1}{12}ML^2$	M = mass (kg); L = length (m); axis through the centre
14	**Thin rod about one end**	$I = \frac{1}{3}ML^2$	M = mass (kg); L = length (m); axis through one end
15	**Radius of gyration**	$I = Mk^2$	k = radius of gyration (m); M = mass (kg); I = moment of inertia (kg m^2)

7.3 Angular Momentum and Rotational Kinetic Energy

16	**Angular momentum**	$L = I\omega = mvr$	L = angular momentum (kg m^2 s^{-1}); I = moment of inertia (kg m^2); ω = angular velocity (rad s^{-1}); mvr applies to a point mass in circular motion

17	**Torque and angular momentum**	$\tau = \frac{dL}{dt} = I\frac{d\omega}{dt} = I\alpha$	τ = torque (N m); α = angular acceleration (rad s^{-2}); I constant
18	**Conservation of angular momentum**	$I_1\omega_1 = I_2\omega_2$	I = moment of inertia (kg m^2); ω = angular velocity (rad s^{-1}); no external torque
19	**Coupling of two rotating bodies**	$I_1\omega_1 + I_2\omega_2 = (I_1 + I_2)\omega$	I_1, I_2 = moments of inertia (kg m^2); ω_1, ω_2 = initial angular velocities (rad s^{-1}); ω = common final angular velocity (rad s^{-1})
20	**Rotational kinetic energy**	$K = \frac{1}{2}I\omega^2 = \frac{L^2}{2I}$	K = rotational kinetic energy (J); I = moment of inertia (kg m^2); ω = angular velocity (rad s^{-1}); L = angular momentum (kg m^2 s^{-1})
21	**Rolling without slipping**	$v = r\omega$; $K = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$	v = speed of centre (m s^{-1}); r = radius (m); ω = angular velocity (rad s^{-1}); K = total kinetic energy (J); m = mass (kg); I = moment of inertia (kg m^2)
22	**Rolling down an incline**	$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$	h = height of incline (m); starts from rest; m = mass (kg); v = speed at the bottom (m s^{-1}); ω = angular velocity (rad s^{-1})
23	**Solid cylinder rolling down an incline**	$v = \sqrt{\frac{4gh}{3}}$	v = speed at the bottom (m s^{-1}); h = height (m); g = gravitational acceleration (m s^{-2}); from rest
24	**Work done by a torque**	$W = \tau\theta$	W = work done (J); τ = torque (N m); θ = angle turned (rad)
25	**Power of a torque**	$P = \tau\omega$	P = power (W); τ = torque (N m); ω = angular velocity (rad s^{-1})

CHAPTER 8 — STATICS

8.1 Equilibrium of Forces

1	**Translational equilibrium**	$\Sigma F = 0$; $\Sigma F_x = 0$ and $\Sigma F_y = 0$	ΣF = resultant force (N); F_x, F_y = components along x and y axes; for coplanar forces
2	**Rotational equilibrium**	$\Sigma\tau = 0$	τ = torque (N m); about any chosen point; Σ clockwise moments = Σ anticlockwise moments
3	**Lami's theorem**	$\frac{F_1}{\sin\alpha} = \frac{F_2}{\sin\beta} = \frac{F_3}{\sin\gamma}$	F_1, F_2, F_3 = three forces in equilibrium (N); α, β, γ = angles opposite F_1, F_2, F_3 (the angles between the other two forces)

4	**Resolving a force**	$F_x = F\cos\theta$; $F_y = F\sin\theta$	F = force (N); θ = angle of F to the horizontal; F_x = horizontal component (N); F_y = vertical component (N)
5	**Weight**	$W = mg$	W = weight (N); m = mass (kg); g = gravitational field strength (N kg^{-1}); acts vertically downwards through the centre of gravity
6	**Friction at a rough surface**	$f \le \mu N$; $f = \mu N$ (limiting)	f = friction force (N); μ = coefficient of friction; N = normal reaction (N); equality holds at the limiting value

8.2 Moments and Centre of Gravity

7	**Moment of a force**	Moment = Fd	F = force (N); d = perpendicular distance to the line of action (m); moment in N m
8	**Principle of moments**	ΣF d (clockwise) = ΣF d (anticlockwise)	F = force (N); d = perpendicular distance (m); about the same point
9	**Centre of gravity of a system of masses**	$\bar{x} = \frac{\Sigma mx}{\Sigma m}$	$\bar{x}$ = position of the centre of gravity (m); m = mass (kg); x = position of mass m (m)
10	**Uniform beam**	$W = mg$	W = weight (N); m = mass (kg); g = gravitational field strength (N kg^{-1}); weight acts at the midpoint
11	**Toppling on a slope**	$\tan\theta > \frac{w}{h}$; $\tan\theta = \frac{\frac{w}{2}}{\frac{h}{2}}$	Uniform block of base width w (m) and height h (m) on a slope of angle θ; topples when $\tan\theta > \frac{w}{h}$; the line through the centre of gravity passes the lower edge at $\tan\theta = \frac{\frac{w}{2}}{\frac{h}{2}}$
12	**Toppling on a horizontal surface**	$P\,h \ge W\,\frac{w}{2}$	P = horizontal force (N) applied at height h (m); W = weight (N); w = base width (m); block topples when this holds
13	**Reaction forces on a beam**	$R_1 + R_2 = W$ + other loads	R_1, R_2 = reactions at the two supports (N); W = weight of beam (N); translational equilibrium

▶ CHAPTER 9 — DEFORMATION OF SOLIDS

9.1 Stress, Strain and Young's Modulus

1	**Stress**	$\sigma = \frac{F}{A}$	σ = stress (Pa = N m^{-2}); F = normal force (N); A = cross-sectional area (m^2)
2	**Strain**	$\varepsilon = \frac{x}{L_0}$	ε = strain (no unit); x = extension (m); L_0 = original length (m)
3	**Young modulus**	$E = \frac{\sigma}{\varepsilon} = FL_0/(Ax)$	E = Young modulus (Pa); E is the gradient of the linear part of a σ–ε graph
4	**Force constant**	$F = kx;\ k = \frac{EA}{L_0}$	k = force constant (N m^{-1}); valid below the limit of proportionality; E = Young modulus (Pa); A = cross-sectional area (m^2); L_0 = original length (m)
5	**Area of a circular wire**	$A = \frac{\pi d^2}{4} = \pi r^2$	d = diameter (m); r = radius (m); A = cross-sectional area (m^2)
6	**Extension of a wire under a given load**	$x \propto \frac{L_0}{d^2}$	x = extension (m); L_0 = original length (m); d = diameter (m)

9.2 Elastic Energy

7	**Elastic potential energy**	$U = \frac{1}{2}Fx = \frac{1}{2}kx^2 = \frac{F^2}{2k}$	U = energy stored (J); F = force (N); x = extension (m); k = force constant (N m^{-1}); valid for the linear region only
8	**Elastic energy in terms of Young modulus**	$U = \frac{1}{2}\frac{EA}{L_0}x^2$	E = Young modulus (Pa); A = cross-sectional area (m^2); L_0 = original length (m); x = extension (m)
9	**Energy density**	$\frac{U}{V} = \frac{1}{2}\sigma\varepsilon = \frac{\sigma^2}{2E} = \frac{1}{2}E\varepsilon^2$	unit J m^{-3}; V = AL_0 = volume (m^3); σ = stress (Pa); ε = strain; E = Young modulus (Pa)
10	**Work done for a non-linear curve**	Work done = area under the F–x graph	count squares or use trapezium; F = force (N); x = extension (m)
11	**Extra energy stored when load increases**	$\Delta U = \frac{1}{2}(F_1 + F_2)(x_2 - x_1)$	for a linear graph; load increases from F_1 to F_2 (N); x_1, x_2 = corresponding extensions (m)

CHAPTER 10 — GASES

10.1 Ideal Gas Equation

1	**Ideal gas equation (moles)**	$pV = nRT$	p = pressure (Pa); V = volume (m^3); n = amount of substance (mol); R = 8.31 J mol^{-1} K^{-1}; T = thermodynamic temperature (K)

2	**Ideal gas equation (molecules)**	$pV = NkT$	N = number of molecules; $k = \frac{R}{N_A} = 1.38\times10^{-23}$ J K^{-1} (Boltzmann constant); T = thermodynamic temperature (K)
3	**Amount of substance**	$n = \frac{N}{N_A} = \frac{m}{M}$	$N_A = 6.02\times10^{23}$ mol^{-1}; m = mass (kg); M = molar mass (kg mol^{-1})
4	**Gas law for fixed mass**	$\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$	for a fixed mass of gas; T in kelvin
5	**Temperature conversion**	$T\,(K) = \theta\,(°C) + 273$	T = thermodynamic temperature (K); θ = Celsius temperature (°C); 273.15 may be used instead of 273
6	**Density of an ideal gas**	$\rho = \frac{pM}{RT}$	from $pV = \frac{m}{M}RT$; ρ = density (kg m^{-3}); M = molar mass (kg mol^{-1})

10.2 Kinetic Theory and Molecular Speeds

7	**Pressure from kinetic theory**	$p = \frac{1}{3}\rho\langle c^2\rangle = \frac{1}{3}(Nm/V)\langle c^2\rangle$	ρ = density (kg m^{-3}); N = number of molecules; m = mass of one molecule (kg); V = volume (m^3); $\langle c^2\rangle$ = mean square speed (m^2 s^{-2})
8	**Root-mean-square speed**	$c_{rms} = \sqrt{\langle c^2\rangle} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3p}{\rho}}$	M = molar mass (kg mol^{-1}); m = mass of one molecule (kg); T = temperature (K); k = Boltzmann constant (J K^{-1}); R = molar gas constant (J mol^{-1} K^{-1}); p = pressure (Pa); ρ = density (kg m^{-3})
9	**Mean translational kinetic energy per molecule**	$\langle E_k\rangle = \frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$	for the translational motion of one molecule (J)
10	**Mean energy per molecule**	$\frac{f}{2}kT$	f = number of degrees of freedom; k = Boltzmann constant (J K^{-1}); T = temperature (K)
11	**Internal energy of an ideal gas**	$U = \frac{f}{2}nRT = \frac{f}{2}NkT$	n = number of moles; N = number of molecules; f = number of degrees of freedom; U in J
12	**Total translational kinetic energy**	$\frac{3}{2}nRT$	n moles of gas; total translational kinetic energy in J; T = temperature (K)
13	**Pressure and kinetic energy**	$pV = \frac{2}{3}N\langle E_k\rangle$	N = number of molecules; $\langle E_k\rangle$ = mean translational kinetic energy per molecule (J)

CHAPTER 11 — THERMODYNAMICS

11.1 First Law of Thermodynamics

1	**First law of thermodynamics**	$\Delta U = Q - W$	ΔU = change in internal energy (J); Q = heat supplied to the gas (J), negative when heat leaves the gas; W = work done by the gas (J), negative when the gas is compressed
2	**Change in internal energy**	$\Delta U = \frac{f}{2} nR\Delta T$	for an ideal gas, whatever the process; f = number of degrees of freedom; n = number of moles; ΔT = temperature change (K)
3	**Work done by gas**	$W = p\Delta V$; W = area under the p–V graph	$W = p\Delta V$ at constant pressure; the area under the p–V graph holds in general; p = pressure (Pa); ΔV = volume change (m^3)
4	**Heat at constant volume**	$Q = nC_V\Delta T$	C_V = molar heat capacity at constant volume (J mol^{-1} K^{-1}); n = number of moles; ΔT = temperature change (K)
5	**Heat at constant pressure**	$Q = nC_p\Delta T$	C_p = molar heat capacity at constant pressure (J mol^{-1} K^{-1}); n = number of moles; ΔT = temperature change (K)
6	**Molar heat capacities**	$C_V = \frac{f}{2}R$; $C_p = C_V + R$	Mayer's relation: $C_p = C_V + R$; R = molar gas constant (J mol^{-1} K^{-1}); f = number of degrees of freedom
7	**Ratio of heat capacities**	$\gamma = \frac{C_p}{C_V} = \frac{f+2}{f}$	γ = ratio of molar heat capacities (no unit); f = number of degrees of freedom

11.2 Thermodynamic Processes

8	**Isothermal process**	$W = nRT \ln\frac{V_2}{V_1} = p_1V_1 \ln\frac{V_2}{V_1}$; pV = constant	W = work done by the gas (J); n = number of moles; T = constant temperature (K); V_1, V_2 = initial and final volumes (m^3); p_1 = initial pressure (Pa)
9	**Adiabatic relations**	pV^γ = constant; $TV^{\gamma-1}$ = constant; $T^\gamma p^{1-\gamma}$ = constant	$\gamma = \frac{C_p}{C_V}$ ($\frac{5}{3}$ monatomic, $\frac{7}{5}$ diatomic)
10	**Adiabatic work**	$W = \frac{p_1V_1 - p_2V_2}{\gamma - 1} = \frac{nR(T_1 - T_2)}{\gamma - 1}$ $= -\Delta U$	W = work done by the gas (J); p_1, V_1, T_1 = initial state; p_2, V_2, T_2 = final state; n = number of moles; R = molar gas constant (J mol^{-1} K^{-1}); $\gamma = \frac{C_p}{C_V}$
11	**Isobaric process**	$W = p\Delta V = nR\Delta T$; $Q = nC_p\Delta T$	constant pressure; p = pressure (Pa); ΔV = volume change (m^3); ΔT = temperature change (K); C_p = molar heat capacity at constant pressure (J mol^{-1} K^{-1})

12	**Isochoric process**	$W = 0;\ Q = \Delta U = nC_V\Delta T$	constant volume; C_V = molar heat capacity at constant volume (J mol^{-1} K^{-1}); ΔT = temperature change (K)
13	**Mayer's relation**	$C_p - C_V = R$	C_p, C_V = molar heat capacities at constant pressure and volume (J mol^{-1} K^{-1}); R = molar gas constant (J mol^{-1} K^{-1})
14	**First law for each process**	$Q = \Delta U + W$	Q = heat supplied to the gas (J); ΔU = change in internal energy (J); W = work done by the gas (J)

CHAPTER 12 — TRANSFER OF THERMAL ENERGY

12.1 Conduction, Convection and Radiation

1	**Rate of heat conduction**	$\frac{Q}{t} = \frac{kA(\theta_1 - \theta_2)}{L} = -kA\frac{d\theta}{dx}$	$\frac{Q}{t}$ = rate of heat flow (W); k = thermal conductivity (W m^{-1} K^{-1}); A = cross-sectional area (m^2); L = length or thickness (m); θ_1, θ_2 = temperatures of the two faces; $\frac{d\theta}{dx}$ = temperature gradient (K m^{-1})
2	**Thermal resistance**	$R = \frac{L}{kA}$	R = thermal resistance (K W^{-1}); L = length or thickness (m); k = thermal conductivity (W m^{-1} K^{-1}); A = cross-sectional area (m^2)
3	**Layers in series**	$R = R_1 + R_2 + \ldots$	R = total thermal resistance (K W^{-1}); R_1, R_2, … = thermal resistances of the layers
4	**Heat flow through thermal resistance**	$\frac{Q}{t} = \frac{\Delta\theta}{R}$	$\Delta\theta$ = temperature difference (K); R = thermal resistance (K W^{-1})
5	**Power radiated**	$P = e\sigma AT^4$	P = power radiated (W); $\sigma = 5.67\times10^{-8}$ W m^{-2} K^{-4}; e = emissivity; A = surface area (m^2); T = temperature (K, in kelvin)
6	**Net power radiated**	$P_{net} = e\sigma A(T^4 - T_0^4)$	body at temperature T in surroundings at temperature T_0 (K); P_{net} = net power radiated (W); e = emissivity; A = surface area (m^2); σ = Stefan–Boltzmann constant
7	**Surface area of a sphere**	$A = 4\pi r^2$	for a spherical star or planet; r = radius (m); A = surface area (m^2)
8	**Intensity at a distance**	$I = \frac{P}{4\pi d^2}$	I = intensity at distance d (W m^{-2}); P = power emitted (W); d = distance from the source (m)